

# Connection-Based Aloha: Modeling, Optimization, and Effects of Connection Establishment

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**Abstract**—According to whether a connection to the receiver is established before data transmission, random access schemes can be divided into two categories: connection-based and connection-free. For cellular networks, its connection-based access mechanism has been heavily criticized for being ill-suited to Machine-to-Machine (M2M) communications. With the existing studies overwhelmingly focusing on the physical-layer transmission and detection design of various connection-free access schemes, less attention has been paid to the fundamental understanding of when connection establishment could be beneficial. The challenge originates from the lack of proper modeling of connection-based random access. In this paper, a scalable model that incorporates each node's queueing behavior is proposed for connection-based slotted Aloha, based on which both the data throughput and the mean queueing delay of data packets are characterized, optimized, and compared with those for connection-free slotted Aloha. Conditions for beneficial connection establishment in terms of the optimal throughput and queueing delay performance are discussed and applied to cellular networks with the newly introduced Random Access-based Small Data Transmission (RA-SDT) schemes. The analysis offers important insights for the optimal access design of M2M communications.

**Index Terms**—Random access, Aloha, connection-based, connection-free, grant-based, grant-free, throughput, queueing delay, machine-to-machine (M2M) communications.

## I. INTRODUCTION

THE rapid proliferation of Machine-Type Devices (MTDs) is yielding a paradigm shift from the traditional Human-to-Human (H2H) communications to Machine-to-Machine (M2M) communications [1]. A central challenge brought by M2M communications is to support massive access of MTDs in an efficient and timely manner. As a fundamental type of multiple access, random access, with which nodes independently decide when to transmit in a distributed fashion, is an appealing solution to M2M communications by virtue of its simplicity and scalability.

Numerous random access protocols have been proposed since Abramson's pioneering work on Aloha [2]. According

to whether a connection is established before the data packet transmission, they can be broadly divided into two categories: connection-free and connection-based. Specifically, with connection-free random access schemes, nodes directly transmit their data packets. When there are concurrent transmissions from multiple nodes, transmission failures may occur, and the resultant wasted time is determined by the length of a data packet. For connection-based random access, on the other hand, each node first sends a request, and transmits its data packets only after receiving the acknowledgement from the receiver that indicates the successful detection of its request, i.e., a connection is established. As a consequence, only requests are involved in transmission failures, and data packets can be transmitted over the reserved resources.

Intuitively, when the data packet length is much larger than the request length, connection-based random access is preferred as the benefit brought by the reduction of transmission failure time would outweigh the extra overhead for connection establishment. Nevertheless, connection establishment may become costly for short-packet transmissions, which is common in M2M communications. Such concerns have long been raised over the Long Term Evolution (LTE) cellular systems, where its grant-based (i.e., connection-based) access mechanism has been intensely criticized for the low efficiency and long delay caused by the connection-establishment procedure [3]. To this end, there have been extensive discussions on the suitability of grant-based random access schemes for M2M communications, with many advocating that grant-free (i.e., connection-free) random access is more favorable [4], [5], [6]. Most of the studies have focused on improving the success probability of uplink data transmission in grant-free random access by designing various transmission schemes for collision reduction [7], [8], [9], [10] and detection algorithms that exploit the sparsity nature of MTDs' activities [11], [12], [13], [14], [15], [16], [17], [18], [19].

Note that the advanced data transmission and detection schemes proposed for grant-free random access can also be applied to grant-based random access to reduce the request transmission failures. Given the same physical-layer design, from the perspective of medium access control layer, whether a grant-free random access protocol outperforms a grant-based one is determined by the data payload length and protocol overhead. Specifically, for given overhead parameters, there exists a critical threshold of data payload length, only above which connection establishment is beneficial. Such a threshold was recently characterized in [20] for both Aloha and Carrier Sense Multiple Access (CSMA) in terms of the maximum data throughput. The analysis shows that for Aloha, the threshold

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may drop below zero if the overhead parameters satisfy a certain constraint, indicating that connection establishment can be beneficial for any data payload length.

There are, nevertheless, a few limitations when applying the above results to practice. First of all, the optimal throughput performance was considered in [20], to achieve which nodes' queues should be pushed to saturation, i.e., the queues are always busy. When the queueing delay performance becomes the main concern, however, the focus should be shifted to the unsaturated condition, only under which the mean queueing delay of data packets is bounded. More importantly, in [20], a channel-centric model was adopted, with which the throughput analysis can be greatly simplified by only focusing on the aggregate traffic of nodes instead of each individual node's behavior. Yet by excluding nodes' queues from the model, it is difficult to further characterize the queueing performance and obtain the optimal setting of access parameters of nodes, which is of paramount importance for practical network control.

With nodes' queues taken into consideration, a node-centric model was developed in [21] for connection-free Aloha, where the coupled service rates of nodes' queues were derived by characterizing the state transition of Head-of-Line (HOL) packets in each queue and establishing the fixed-point equations of the steady-state probability of successful transmission of HOL packets in saturated and unsaturated conditions. The model in [21], however, cannot be directly applied to connection-based Aloha because instead of contending for channel access individually, with connection-based Aloha, data packets are transmitted in a batch after a connection is established. That calls for a novel node-centric model for connection-based Aloha, based on which both throughput and queueing delay performance can be characterized, optimized, and further compared with that of connection-free Aloha.

Note that for connection-based random access, the requests, rather than data packets, contend with each other. Therefore, the key to performance analysis lies in proper modeling of the behavior of each node's requests. In this paper, a request-queue model is proposed for throughput and delay optimization of connection-based slotted Aloha. Specifically, by introducing a request queue for each node, both the data throughput and the mean queueing delay of data packets are derived based on the service time distribution of the request queue. The optimal transmission probability of nodes for maximizing the data throughput or minimizing the mean queueing delay is also obtained as an explicit function of the number of nodes or the data input rate of each node. The optimal throughput and delay performance of connection-based slotted Aloha is further compared with that of connection-free slotted Aloha, based on which the criteria for beneficial connection establishment are discussed. The analysis is also applied to the Fifth-Generation (5G) cellular networks for performance comparison of the recently introduced grant-free 2-step Random Access-based Small Data Transmission (RA-SDT) and grant-based 4-step RA-SDT schemes [22].

The remainder of this paper is organized as follows. Related work is reviewed in Section II. Section III describes our system model of connection-based slotted Aloha. In Section IV, the service process of each request queue is characterized, based

on which the maximum data throughput and the minimum mean queueing delay are derived and verified by simulation results in Section V. The comparative analysis is conducted between connection-free slotted Aloha and connection-based slotted Aloha and further applied to 5G networks in Section VI. Possible extensions of the analysis are discussed in Section VII. Finally, conclusions are summarized in Section VIII.

## II. RELATED WORK

In this section, we will review the previous studies on modeling and performance analysis of connection-based Aloha.

Similar to connection-free Aloha, models for connection-based Aloha can be in general categorized into channel-centric and node-centric, with the former focusing on the aggregate requests and the latter on each node's queueing behavior. Early studies adopted channel-centric models to analyze the throughput and access delay performance by either assuming a network with an infinite number of nodes [23], [24], or a finite-node network with each node equipped of a one-packet buffer [25], [26], [27]. The number of aggregate requests of nodes is usually assumed as Poisson distributed [23], [24], or characterized based on the state transition of the number of backlogged nodes (i.e., nodes with nonempty buffer) [25], [26], [27]. The proposed channel-centric models have been widely adopted in later studies on modeling and performance optimization for connection-based access schemes in LTE and 5G cellular systems [28], [29], [30], [31], [32], [33]. When modeling a cellular network with massive MTDs, for instance, the number of requests per time slot is often assumed to follow a Poisson distribution [28], [29]. However, by ignoring each node's data queue, the adaptive tuning of access parameters, such as the transmission probability of each node, can only be performed based on the time-varying network status such as the number of backlogged nodes [30], [31], [32], [33], which in practice is difficult to track and estimate accurately, rather than the data input rate of each node.

To include nodes' queues, various node-centric models have been proposed in [34], [35], and [36]. In [34], the joint distribution of nodes' queue lengths was obtained for a two-node connection-based slotted Aloha network, which is, nevertheless, difficult to be generalized to an  $n$ -node network due to high modeling complexity. Therefore, the symmetric scenario, where nodes have identical data input rates and transmission probabilities, has been considered in [35] and [36], where the modeling dimensions can be greatly reduced by focusing on each single queue. Specifically, in [35], each node's data buffer was modeled as a  $Geo/D/1$  queue with vacation. By assuming that the number of aggregate requests in each time slot is Poisson distributed with the average rate  $G$ , the mean queueing delay of data packets was derived as a function of  $G$ . Nevertheless, the assumption of Poisson distributed number of requests holds only when the number of nodes goes to infinity and the transmission probability of each request approaches zero. In [36], the contention among requests was decoupled from the queueing analysis of data packets by establishing a Markov process for the requests. The network throughput was further characterized and maximized

by optimally tuning the transmission probability of each node based on the aggregate data input rate.

Note that the model proposed in [36] was customized for the LTE cellular systems, where some key features, such as clearing the data buffer after connection establishment, are no longer adopted in the RA-SDT schemes of 5G systems.<sup>1</sup> As specified in the newest standard [22], small data payload of a fixed length can be transmitted after an uplink grant is obtained, with which the connection-establishment procedure is simplified without the need for reporting the buffer length. In this paper, instead of focusing on a specific network, we will develop a general analytical model for connection-based slotted Aloha with the fixed-length data transmission incorporated. RA-SDT of 5G cellular systems will be included as a case study.

### III. SYSTEM MODEL

Consider a slotted Aloha network where  $n$  nodes transmit to a single receiver, each with an identical arrival process of data packets and equipped with an infinite-size buffer to store incoming packets. All the nodes are synchronized and can start a transmission only at the beginning of a time slot.<sup>2</sup> Assume the collision model at the receiver, that is, if more than one node transmits simultaneously, a collision occurs and all of them fail. A transmission is successful if and only if there are no concurrent transmissions.

With connection-based slotted Aloha, nodes would send requests to establish a connection before their data transmissions. Assume that each node sends a request with probability  $q$  when a batch of  $M$  data packets is formed in the buffer. If a request is successfully received, i.e., an acknowledgement (ACK) is sent by the receiver, then the node can start transmitting  $M$  data packets. Otherwise, it retransmits the request with probability  $q$  in the next time slot.<sup>3</sup>

As Fig. 1 illustrates, each successful transmission includes  $M$  data packets and the signaling overhead. Without loss of generality, suppose that the transmission time of each packet is equal to one time slot. The overhead time of each successful transmission is denoted as  $\delta$ , in the unit of time slots, where one time slot is for connection establishment and the other

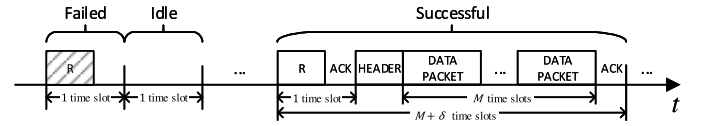


Fig. 1. Graphic illustration of connection-based slotted Aloha.

$\delta - 1$  time slots are for signaling exchange (e.g., header and ACK) during the data transmission.

In this paper, we are interested in throughput and delay performance of connection-based slotted Aloha networks. For each node, as at most one data packet is successfully transmitted in each time slot, the average number of successfully transmitted data packets per time slot is equal to the fraction of time for its successful data packet transmission, which is defined as its data throughput  $\lambda_{out}$ . The data throughput of the network is then  $\hat{\lambda}_{out} = n\lambda_{out}$ .

For each data packet, define its queueing delay  $T$  as the time from its arrival till its successful transmission, in the unit of time slots. To be more specific, the queueing delay of a data packet consists of the waiting time in the buffer and the service time. The waiting time in the buffer includes the time for forming a batch with  $M$  data packets, and the time for this batch waiting to be the HOL batch. The service time of a data packet includes the time of its batch contending for channel access, and the transmission time after a connection is successfully established.

Both data throughput performance and queueing delay performance of each node crucially depend on whether the node's data queue is saturated or not. Denote the input rate (i.e., the average number of arrival data packets per time slot) and the service rate (i.e., the average number of data packets that can be successfully transmitted per time slot given that the data queue is nonempty) of each node's data queue as  $\lambda$  and  $\mu$ , respectively. When  $\lambda < \mu$ , the data queue is unsaturated with a non-zero probability of being empty. If  $\lambda \geq \mu$ , then the data queue is saturated. The data throughput of each node  $\lambda_{out}$  can be written as

$$\lambda_{out} = \begin{cases} \lambda & \lambda < \mu \\ \mu & \lambda \geq \mu. \end{cases} \quad (1)$$

For the mean queueing delay of data packets  $\bar{T}$ , we have  $\bar{T} < \infty$  when the data queue is unsaturated, and  $\bar{T} = \infty$  if the data queue becomes saturated.

It can be seen that for each node's data queue, to characterize the throughput and delay performance, the key lies in the characterization of service rate, which is determined by the transmission probability  $q$  of requests and the probability of successful transmission of requests. In Section IV, a request-queue model will be proposed to characterize the service rate  $\mu$ , based on which the data throughput of the network and the mean queueing delay of data packets will further be obtained and optimized by properly choosing the transmission probability  $q$  in Section V. Specifically, define the maximum data throughput as

$$\hat{\lambda}_{max} = \max_{q, \lambda} \hat{\lambda}_{out}, \quad (2)$$

<sup>1</sup>Note that a similar assumption was adopted in [35], that is, a node connected to the receiver would clear its data buffer if the number of data packets in the buffer is no more than an upper limit. The optimal upper limit for minimizing the mean queueing delay of data packets was shown to be infinity, with which each node always clears its buffer after establishing a connection.

<sup>2</sup>Note that synchronization among nodes is crucial for network performance. Intuitively, if nodes are not synchronized, more transmission failures would occur, leading to lower efficiency. In practice, nodes acquire time synchronization before their transmissions. In the 5G networks, for instance, each node needs to first detect the synchronization signals that are periodically broadcast by the base station before the initial access [37]. Based on the synchronization signals, nodes can identify the radio subframe boundary, with which their transmissions can then be synchronized.

<sup>3</sup>Note that in the literature, there have been various assumptions on the transmission probability of each node. For instance, it can be adaptively adjusted based on the number of transmission failures a node has experienced [38], or the queue status of other nodes [39]. In this paper, the transmission probability is assumed to be independent of each node's prior experience or queue status. It, nevertheless, can be tuned based on the total number of nodes and each node's input rate of data packets to optimize the throughput and delay performance, as will be shown later.

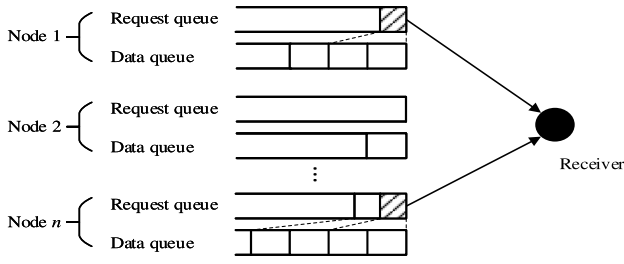


Fig. 2. Illustration of request-queue model.

where  $\hat{\lambda} = n\lambda$  is the aggregate input rate of data packets. For the mean queueing delay of data packets  $\bar{T}$ , as it is finite only in the unsaturated condition, we need to first find out how to select the transmission probability  $q$  such that the data queue is unsaturated. Define the unsaturated region of transmission probability  $q$  as

$$S_{u^n}(q, \hat{\lambda}) = \{q : \hat{\lambda} < \hat{\mu}\}, \quad (3)$$

where  $\hat{\lambda} = n\lambda$  and  $\hat{\mu} = n\mu$  are the aggregate input rate and the aggregate service rate of data packets, respectively. For a given aggregate input rate  $\hat{\lambda}$ , the minimum mean queueing delay is defined as

$$\bar{T}^* = \min_{q \in S_{u^n}(q, \hat{\lambda})} \bar{T}. \quad (4)$$

Expressions of the maximum data throughput  $\hat{\lambda}_{\max}$  and the minimum mean queueing delay  $\bar{T}^*$  with connection-based slotted Aloha will be derived in Section V, and compared to those with connection-free slotted Aloha in Section VI.

#### IV. REQUEST-QUEUE MODEL

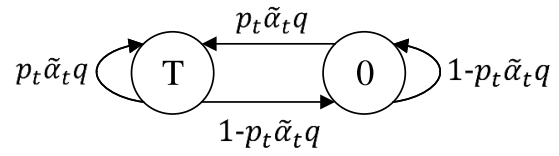
For connection-based slotted Aloha, data packets are served in a batch of  $M$  data packets. To analyze the service process of data packets, in this section, we propose a request-queue model for each node.

As Fig. 2 illustrates, for the virtual request queue of each node, a request is generated upon the arrival of  $M$  data packets, and removed from the request queue after this batch of  $M$  data packets is successfully transmitted. Therefore, the number of requests in a node's request queue is indeed the number of data-packet batches in its data queue. The input rate  $\lambda_r$  and the service rate  $\mu_r$  of each request queue can then be written as

$$\lambda_r = \frac{\lambda}{M}, \quad \mu_r = \frac{\mu}{M}, \quad (5)$$

where  $\lambda$  and  $\mu$  are the input rate and the service rate of each data queue, respectively.

Note that the request queues of nodes are coupled with each other through the service processes. In the following subsections, we will derive the service rate of each request queue by characterizing the state transition of HOL requests and establishing the fixed-point equation of the probability of successful transmission of HOL requests.

Fig. 3. Embedded Markov chain  $\{X_j\}$  of the state transition process of an individual HOL request in connection-based slotted Aloha networks.

##### A. State Characterization of HOL Requests

The behavior of each HOL request can be characterized by a discrete-time Markov renewal process  $(\mathbf{X}, \mathbf{V}) = \{(X_j, V_j), j = 0, 1, \dots\}$ , where  $X_j$  denotes the state of a tagged HOL request at the  $j$ -th transition and  $V_j$  denotes the epoch at which the  $j$ -th transition occurs. The embedded Markov chain  $\mathbf{X} = \{X_j\}$  is shown in Fig. 3.

As shown in Fig. 3, a HOL request is initially in State T, and stays in State T if there is no on-going data transmission and the request is successfully transmitted. It moves to State 0 if the channel is occupied by an on-going data transmission, or there is no data transmission but 1) the request is not transmitted or 2) the request is transmitted but fails. For a given HOL request, let  $\tilde{\alpha}_t$  denote the probability that there is no data transmission at time slot  $t = 1, 2, \dots$ , and  $p_t$  denote the probability of successful transmission conditioned on no data transmission at time slot  $t = 1, 2, \dots$ . The steady-state probability distribution of the embedded Markov chain in Fig. 3 can be easily obtained as

$$\pi_T = p\tilde{\alpha}q, \quad \pi_0 = 1 - p\tilde{\alpha}q, \quad (6)$$

where  $\tilde{\alpha} = \lim_{t \rightarrow \infty} \tilde{\alpha}_t$  and  $p = \lim_{t \rightarrow \infty} p_t$  are the steady-state probability of no data transmission and the steady-state probability of successful transmission conditioned on no data transmission, respectively, for a given HOL request.

Denote the mean holding time at State T and State 0 as  $\tau_T$  and  $\tau_0$ , respectively, which are given by

$$\tau_T = M + \delta, \quad \tau_0 = 1, \quad (7)$$

in the unit of time slots. The limiting state probabilities of the Markov renewal process  $(\mathbf{X}, \mathbf{V})$  can be then written as

$$\tilde{\pi}_i = \frac{\pi_i \tau_i}{\sum_{j \in \{T, 0\}} \pi_j \tau_j}, \quad i \in \{T, 0\}. \quad (8)$$

By combining (6)-(8), we have

$$\tilde{\pi}_T = \frac{p\tilde{\alpha}q(M+\delta)}{p\tilde{\alpha}q(M+\delta-1)+1}, \quad \tilde{\pi}_0 = \frac{1-p\tilde{\alpha}q}{p\tilde{\alpha}q(M+\delta-1)+1}. \quad (9)$$

As a successful transmission of each HOL request only occurs at State T and holds for  $\tau_T$  time slots, the service rate  $\mu_r$  of HOL requests is given by  $\frac{\tilde{\pi}_T}{\tau_T}$ , which can be further written from (7) and (9) as

$$\mu_r = \frac{p\tilde{\alpha}q}{p\tilde{\alpha}q(M+\delta-1)+1}. \quad (10)$$

Note from (10) that the service rate  $\mu_r$  is determined by  $\tilde{\alpha}$  and  $p$ , which depend on whether the request queue is saturated or not. The steady-state probability of no data transmission for a given HOL request  $\tilde{\alpha}$  can be obtained as

$$\tilde{\alpha} = \begin{cases} \frac{\alpha}{1-\lambda_r(M+\delta-1)} & \lambda_r < \mu_r \\ \frac{\alpha}{1-p\alpha q(M+\delta-1)} & \lambda_r \geq \mu_r, \end{cases} \quad (11)$$

where  $\alpha$  denotes the unconditional steady-state probability of no data transmission (i.e., regardless of whether there are HOL requests or not):

$$\alpha = \begin{cases} 1 - n\lambda_r(M + \delta - 1) & \lambda_r < \mu_r \\ \frac{1}{nqp(M + \delta - 1) + 1} & \lambda_r \geq \mu_r. \end{cases} \quad (12)$$

Appendix A presents the detailed derivation of (11)-(12). The steady-state probability of successful transmission of HOL requests conditioned on no data transmission  $p$  will be characterized in the following section.

### B. Probability of Successful Transmission of HOL Requests

Whether a HOL request can be successfully transmitted is determined by the activities of HOL requests in other nodes' request queues. Specifically, given that there is no on-going data transmission, a node can successfully transmit its request if and only if all the other  $n - 1$  nodes do not request. Consequently, for a given HOL request, the steady-state probability of successful transmission conditioned on no data transmission,  $p$ , can be written as

$$p = (1 - \Pr\{\text{a request queue is nonempty and the HOL request is transmitted} \mid \text{no data transmission}\})^{n-1}. \quad (13)$$

Let  $\rho$  denote the probability of a request queue being nonempty, which is given by

$$\rho = \begin{cases} \frac{\lambda_r}{\mu_r} & \lambda_r < \mu_r \\ 1 & \lambda_r \geq \mu_r. \end{cases} \quad (14)$$

Note that the transmission probability of each HOL request is  $q$ . Moreover, given that there is no on-going data transmission, each request is either empty, or nonempty with the HOL request at State 0 or the first time slot of State T. (13) can then be written as

$$p = \left(1 - \frac{\rho(\tilde{\pi}_0 + \frac{\tilde{\pi}_T}{\tau_T})}{1 - \rho + \rho(\tilde{\pi}_0 + \frac{\tilde{\pi}_T}{\tau_T})} \cdot q\right)^{n-1}. \quad (15)$$

By combining (7), (9)-(11), and (14)-(15), we have

$$p = \begin{cases} (1 - \frac{\lambda_r}{p\alpha})^{n-1} \approx \exp\{-\frac{n\lambda_r}{p\alpha}\} & \lambda_r < \mu_r \\ (1 - q)^{n-1} \approx \exp\{-nq\} & \lambda_r \geq \mu_r, \end{cases} \quad (16)$$

where the approximation is based on  $n-1 \approx n$  and  $(1-x)^n \approx e^{-nx}$  for  $0 < x < 1$  if  $n \gg 1$ .

For the unsaturated condition, the fixed-point equation of  $p$  given in (16) can be further written as

$$p = \exp\left\{-\frac{n\lambda_r}{p[1 - n\lambda_r(M + \delta - 1)]}\right\}, \quad (17)$$

by combining with (12). Note that (17) has two positive real roots:

$$p_L = \exp\left\{\mathbb{W}_0\left(-\frac{n\lambda_r}{1 - n\lambda_r(M + \delta - 1)}\right)\right\}, \quad (18)$$

and

$$p_S = \exp\left\{\mathbb{W}_{-1}\left(-\frac{n\lambda_r}{1 - n\lambda_r(M + \delta - 1)}\right)\right\}, \quad (19)$$

if and only if

$$\frac{n\lambda_r}{1 - n\lambda_r(M + \delta - 1)} < e^{-1}, \quad (20)$$

where  $\mathbb{W}_0(\cdot)$  and  $\mathbb{W}_{-1}(\cdot)$  are two branches of the Lambert W function [40]. It can be further shown that for the two fixed points  $p_L$  and  $p_S$  of (17),  $p_L$  is an attracting fixed point, while  $p_S$  is a repelling fixed point. Therefore, only the larger root  $p_L$  is the steady-state point in the unsaturated condition. The steady-state point in the saturated condition, denoted as  $p_A$ , can also be obtained from (16) as

$$p_A = e^{-nq}. \quad (21)$$

It can be observed from (18) and (21) that the unsaturated steady-state point  $p_L$  is determined by the aggregate input rate  $\hat{\lambda}$ , the batch size  $M$  and the overhead time for each successful transmission  $\delta$ . In contrast, the saturated steady-state point  $p_A$  is determined by the transmission probability  $q$  and the number of nodes  $n$ .

### C. Service Rate of HOL Requests

It has been shown in (11) and (16) that both  $\tilde{\alpha}$  and  $p$  depend on whether the request queue is saturated or not. Let  $\mu_r^u$  and  $\mu_r^s$  denote the service rate in the unsaturated condition and saturated condition, respectively, which can be written as

$$\mu_r^u = \frac{p_L q [1 - n\lambda_r(M + \delta - 1)]}{p_L q(M + \delta - 1)[1 - n\lambda_r(M + \delta - 1)] - \lambda_r(M + \delta - 1) + 1} \quad (22)$$

and

$$\mu_r^s = \frac{p_A q}{np_A q(M + \delta - 1) + 1}, \quad (23)$$

by combining (10)-(12). Further note that for the request queue to stay unsaturated, the input rate  $\lambda_r$  should not exceed the service rate  $\mu_r^s$ , i.e.,

$$\lambda_r < \frac{qe^{-nq}}{nqe^{-nq}(M + \delta - 1) + 1}. \quad (24)$$

Note that (24) can also be written as  $-\frac{1}{n} \ln p_L < q < -\frac{1}{n} \ln p_S$ , where  $p_L$  and  $p_S$  are given in (18) and (19), respectively. By combining (3), (5), and (18)-(19), the unsaturated region of transmission probability  $q$  can be obtained as

$$S_{u^n}(q, \hat{\lambda}) = \left\{q : -\frac{1}{n} \mathbb{W}_0\left(-\frac{\hat{\lambda}}{M - \hat{\lambda}(M + \delta - 1)}\right) < q < -\frac{1}{n} \mathbb{W}_{-1}\left(-\frac{\hat{\lambda}}{M - \hat{\lambda}(M + \delta - 1)}\right)\right\}. \quad (25)$$

Finally, by combining (22)-(23) and (18), (21), (25), the service rate of each request queue  $\mu_r$  can be obtained in (26), as shown at the bottom of the next page.

## V. THROUGHPUT AND DELAY OPTIMIZATION

Based on the request-queue model, in this section, we will derive the data throughput and the mean queueing delay of data packets, and further demonstrate how to optimize them by tuning the transmission probability of nodes.

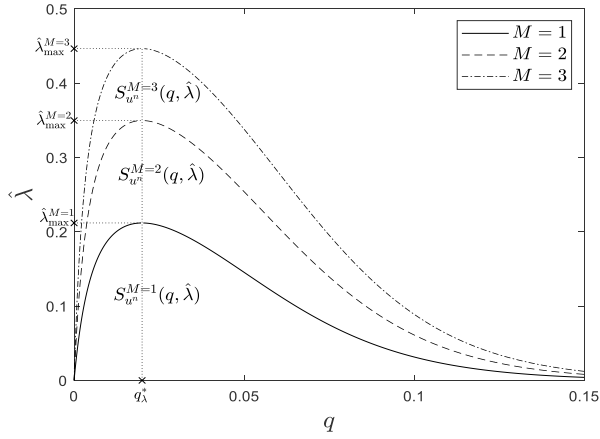


Fig. 4. Unsaturated region  $S_{u^n}(q, \hat{\lambda})$  of transmission probability  $q$ .  $n = 50$ .  $\delta = 2$ .  $M \in \{1, 2, 3\}$ .

#### A. Maximum Data Throughput

By combining (1), (5), and (26), the data throughput  $\hat{\lambda}_{out}$  can be obtained as

$$\hat{\lambda}_{out} = n\lambda_{out} = \begin{cases} \hat{\lambda} & q \in S_{u^n}(q, \hat{\lambda}) \\ \frac{M}{\frac{1}{nqe^{-nq}} + M + \delta - 1} & \text{otherwise.} \end{cases} \quad (27)$$

Note from (25) that  $S_{u^n}(q, \hat{\lambda}) \neq \emptyset$  if and only if (20) holds, which can be further written as

$$\hat{\lambda} < \frac{M}{M + \delta + e - 1}. \quad (28)$$

When  $\hat{\lambda} \geq \frac{M}{M + \delta + e - 1}$ , we have  $S_{u^n}(q, \hat{\lambda}) = \emptyset$  and all the request queues are saturated no matter what value of transmission probability  $q$  is selected. As (27) shows, the data throughput  $\hat{\lambda}_{out}$  in the saturated condition is a function of transmission probability  $q$  and can be maximized as

$$\max_q \frac{M}{\frac{1}{nqe^{-nq}} + M + \delta - 1} = \frac{M}{M + \delta + e - 1} \quad (29)$$

at  $q = \frac{1}{n}$ . Combining (2) with (27)-(29), we then obtain the maximum data throughput  $\hat{\lambda}_{max}$  as

$$\hat{\lambda}_{max} = \frac{M}{M + \delta + e - 1}, \quad (30)$$

which is achieved when the aggregate input rate satisfies  $\hat{\lambda} \geq \frac{M}{M + \delta + e - 1}$  and the transmission probability  $q$  is set to

$$q = q_\lambda^* = \frac{1}{n}. \quad (31)$$

Fig. 4 illustrates the unsaturated region  $S_{u^n}(q, \hat{\lambda})$  and the maximum data throughput  $\hat{\lambda}_{max}$  with various values of batch size  $M$ . When the aggregate input rate  $\hat{\lambda} < \hat{\lambda}_{max}$ ,  $S_{u^n}(q, \hat{\lambda}) \neq \emptyset$ , and each request queue is unsaturated if the transmission probability  $q$  is selected from the unsaturated region. On the other hand, when  $\hat{\lambda} \geq \hat{\lambda}_{max}$ ,  $S_{u^n}(q, \hat{\lambda}) = \emptyset$ , indicating that

each request queue is always saturated for any  $q$ ; the maximum data throughput  $\hat{\lambda}_{max}$  can be achieved at  $q = q_\lambda^*$ . It can be seen from Fig. 4 that an increase in batch size  $M$  leads to both an enlarged unsaturated region and a higher maximum data throughput.

#### B. Minimum Mean Queueing Delay

The queueing delay  $T$  of each data packet consists of three parts: 1) waiting time for forming a batch  $W_b$ ; 2) waiting time of a batch  $W_r$ ; and 3) service time of a batch  $D_r$ . Assuming that data packet arrivals follow a Bernoulli process with parameter  $\lambda$ , we will derive the mean of each part of delay in the following subsections.

1) *Waiting Time for Forming a Batch  $W_b$* : For the  $j$ -th data packet ( $j = 1, 2, \dots, M$ ) in a batch, let  $W_b^{(j)}$  denote its waiting time for forming the batch. With Bernoulli arrivals of data packets, the probability mass function (PMF) of  $W_b^{(j)}$  can be written as

$$\Pr\{W_b^{(j)} = i\} = \begin{cases} \binom{i-1}{M-j-1} \lambda^{M-j} (1-\lambda)^{i-M+j} & i = M-j, M-j+1, \dots \\ 0 & \text{otherwise} \end{cases} \quad (32)$$

for  $j = 1, 2, \dots, M-1$ , and

$$\Pr\{W_b^{(M)} = i\} = \begin{cases} 1 & i = 0 \\ 0 & \text{otherwise.} \end{cases} \quad (33)$$

Note that the probability of a data packet being the  $j$ -th ( $j = 1, 2, \dots, M$ ) one in the batch is equal to  $\frac{1}{M}$ . The mean waiting time for forming a batch  $\bar{W}_b$  can then be obtained as

$$\bar{W}_b = \sum_{j=1}^M \frac{1}{M} \sum_{i=0}^{\infty} i \cdot \Pr\{W_b^{(j)} = i\} = \frac{M-1}{2\lambda}. \quad (34)$$

2) *Service Time of a Batch  $D_r$* :  $D_r$  is also the service time of a HOL request, with the PMF

$$\Pr\{D_r = i\} = \begin{cases} p\tilde{\alpha}q(1-p\tilde{\alpha}q)^{i-M-\delta} & i = M + \delta, M + \delta + 1, \dots \\ 0 & \text{otherwise} \end{cases} \quad (35)$$

according to Fig. 3, where  $\tilde{\alpha}$  is given by (11),  $p = p_L$  in the unsaturated condition is given by (18), and  $p = p_A$  in the saturated condition is given by (21). The mean service time of a batch,  $\bar{D}_r$ , can then be obtained from (35) as

$$\bar{D}_r = \sum_{i=0}^{\infty} i \cdot \Pr\{D_r = i\} = \frac{1}{p\tilde{\alpha}q} + M + \delta - 1. \quad (36)$$

$$\mu_r = \begin{cases} \frac{q \cdot \exp\left\{\mathbb{W}_0\left(-\frac{\hat{\lambda}}{M - \hat{\lambda}(M + \delta - 1)}\right)\right\} \cdot \left[1 - \frac{\hat{\lambda}}{M}(M + \delta - 1)\right]}{q \cdot \exp\left\{\mathbb{W}_0\left(-\frac{\hat{\lambda}}{M - \hat{\lambda}(M + \delta - 1)}\right)\right\} \cdot (M + \delta - 1) \cdot \left[1 - \frac{\hat{\lambda}}{M}(M + \delta - 1)\right] - \frac{\hat{\lambda}}{M}(M + \delta - 1) + 1} & q \in S_{u^n}(q, \hat{\lambda}) \\ \frac{qe^{-nq}}{nqe^{-nq} \cdot (M + \delta - 1) + 1} & \text{otherwise.} \end{cases} \quad (26)$$

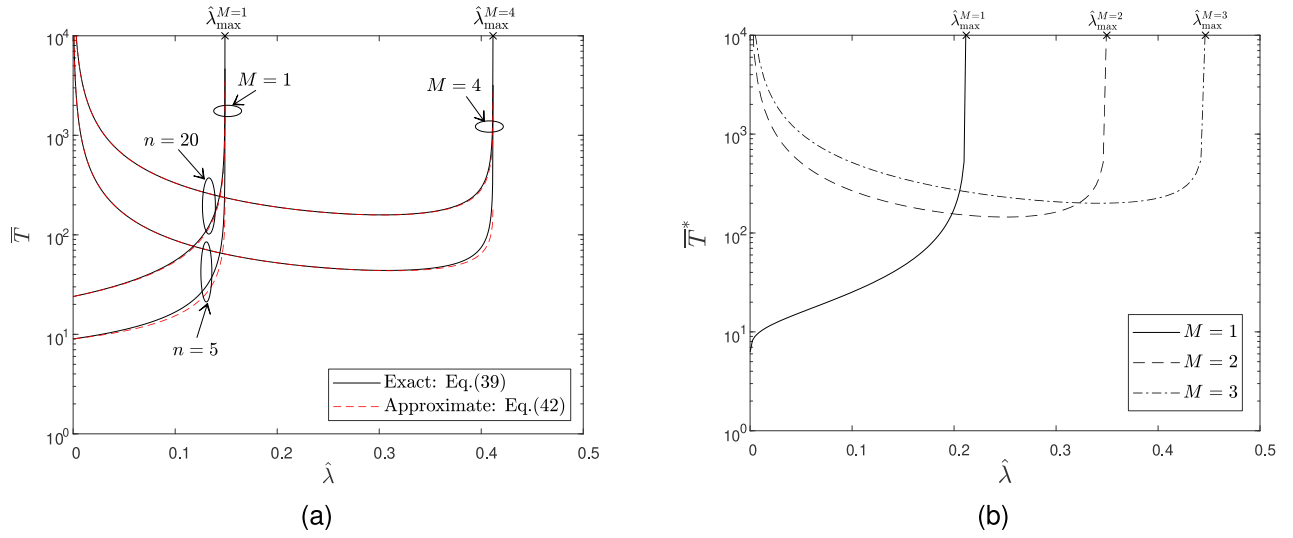


Fig. 5. (a) Exact and approximate expressions of mean queueing delay  $\bar{T}$  versus aggregate input rate  $\hat{\lambda}$ .  $q = \frac{1}{n}$ .  $\delta = 4$ .  $n \in \{5, 20\}$ .  $M \in \{1, 4\}$ . (b) Minimum mean queueing delay  $\bar{T}^*$  versus aggregate input rate  $\hat{\lambda}$ .  $n = 50$ .  $q = q_T^*$ .  $\delta = 2$ .  $M \in \{1, 2, 3\}$ .

3) *Waiting Time of a Batch  $W_r$* :  $W_r$  is indeed the waiting time of a request in the request queue. With Bernoulli arrivals of data packets, we can model each request queue as an  $NB_k/G/1$  queue where the interarrival time of requests follows a negative binomial distribution with parameters  $\lambda$  and  $k = M$ , and the service time of requests follows the distribution given by (35). The mean waiting time  $\bar{W}_r$  is finite only in the unsaturated condition, which can be derived as

$$\bar{W}_r = \frac{1}{\lambda} \sum_{i=1}^{\delta} \frac{r_i}{1-r_i} + 1 - \frac{1}{p_L \tilde{\alpha}_u q}, \quad (37)$$

where  $r_1, r_2, \dots, r_\delta$  are  $\delta$  roots inside the unit circle  $|r| = 1$  of the following equation of  $r$ :

$$p_L \tilde{\alpha}_u q \cdot [(1-\lambda)r + \lambda]^{M+\delta} - (\lambda + p_L \tilde{\alpha}_u q - \lambda p_L \tilde{\alpha}_u q) \cdot r^\delta + (\lambda - \lambda p_L \tilde{\alpha}_u q) \cdot r^{\delta-1} = 0, \quad (38)$$

where  $p_L$  is the unsaturated steady-state point given by (18), and  $\tilde{\alpha}_u = \frac{1-n\lambda_r(M+\delta-1)}{1-\lambda_r(M+\delta-1)}$  is the steady-state probability of no data transmission for a given HOL request in the unsaturated condition according to (11)-(12). Appendix B shows the detailed derivation of (37)-(38).

Finally, by combining (34), (36), and (37), the mean queueing delay of data packets in the unsaturated condition can be obtained as

$$\bar{T} = \frac{M-1}{2\lambda} + M + \delta + \frac{1}{\lambda} \sum_{i=1}^{\delta} \frac{r_i}{1-r_i}, \quad (39)$$

where  $r_1, r_2, \dots, r_\delta$  are  $\delta$  roots inside the unit circle  $|r| = 1$  of Eq. (38) of  $r$ .

Note that the mean queueing delay  $\bar{T}$  is closely dependent on the transmission probability  $q$ . For  $q \notin S_{un}(q, \hat{\lambda})$ , the data queue becomes saturated, leading to  $\bar{T} \rightarrow \infty$ . For  $q \in S_{un}(q, \hat{\lambda})$ , the transmission probability  $q$  should be carefully selected to minimize the mean queueing delay  $\bar{T}$ . Due to the lack of explicit expressions of  $\{r_i\}$ , however, it is difficult to obtain the optimal transmission probability for optimizing  $\bar{T}$

based on (39). In the following, let us introduce an accurate approximation for the mean queueing delay  $\bar{T}$ . Specifically, by approximating the service time distribution of each request as the Geometric distribution with parameter  $p_L \tilde{\alpha}_u q$ , the mean waiting time of each request  $\bar{W}_r$  can be approximated as

$$\bar{W}_r \approx \frac{r_0}{\lambda(1-r_0)} + 1 - \frac{1}{p_L \tilde{\alpha}_u q}, \quad (40)$$

where  $r_0$  is the real root in  $(0, 1)$  of the following equation of  $r$ :

$$(1-\lambda) \cdot r^{M+1} + \lambda \cdot r^M - [\lambda(\frac{1}{p_L \tilde{\alpha}_u q} - 1) + 1] \cdot r + \lambda(\frac{1}{p_L \tilde{\alpha}_u q} - 1) = 0. \quad (41)$$

The derivation of (40)-(41) is presented in Appendix B. We should point out that the mean waiting time of the approximate  $NB_k/Geo/1$  queue is shorter than the actual mean waiting time of each request since we ignore  $M + \delta - 1$  time slots in each service. By replacing  $\bar{W}_r$  with the approximation given in (40), the mean queueing delay  $\bar{T}$  can be approximated as

$$\bar{T} \approx \frac{M-1}{2\lambda} + M + \delta + \frac{r_0}{\lambda(1-r_0)}, \quad (42)$$

where  $r_0$  is the real root in  $(0, 1)$  of Eq. (41) of  $r$ .

The accuracy of (42) is illustrated in Fig. 5a. It is further shown in Appendix C that the mean queueing delay of data packets  $\bar{T}$  given in (42) is a monotonic decreasing function of  $q \in S_{un}(q, \hat{\lambda})$ . Therefore, to minimize  $\bar{T}$ , the optimal transmission probability  $q_T^*$  should be

$$q_T^* = -\frac{1}{n} \mathbb{W}_{-1} \left( -\frac{\hat{\lambda}}{M-\hat{\lambda}(M+\delta-1)} \right) - \epsilon, \quad (43)$$

where  $\epsilon$  is a positive small value.

The minimum mean queueing delay  $\bar{T}^* = \min_{q \in S_{un}(q, \hat{\lambda})} \bar{T}$  can then be obtained by combining (41)-(43). Fig. 5b illustrates how  $\bar{T}^*$  varies with the aggregate input rate  $\hat{\lambda}$  under different values of batch size  $M$ . When  $M = 1$ , for instance, the queueing delay of a data packet only depends on the waiting time and the service time of the request associated to it; in this

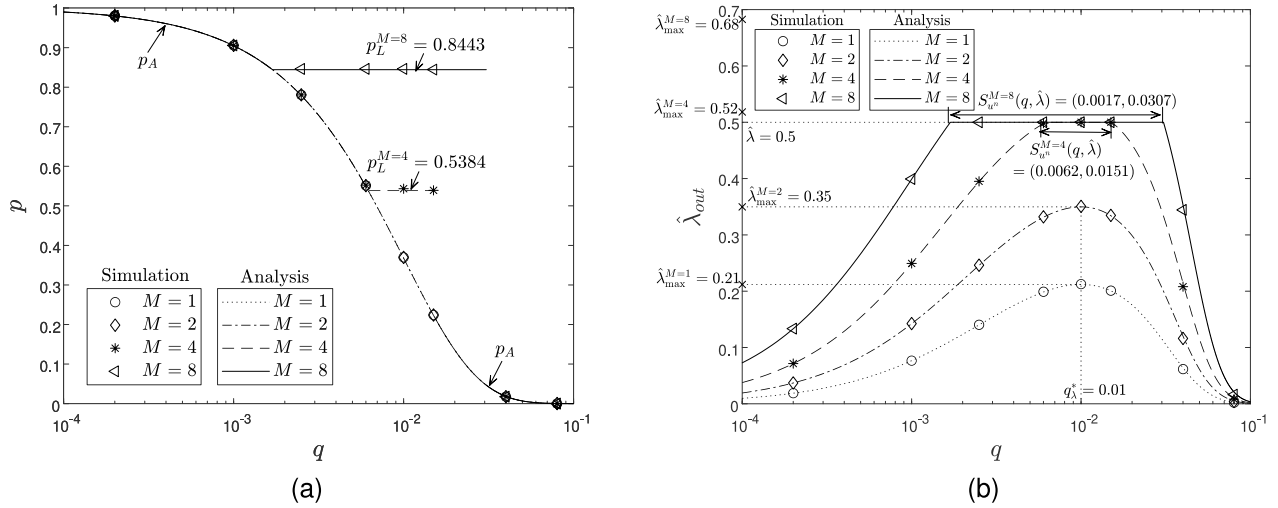


Fig. 6. (a) Steady-state probability of successful transmission of HOL requests conditioned on no data transmission  $p$  and (b) data throughput  $\hat{\lambda}_{out}$  versus transmission probability  $q$ .  $n = 100$ .  $\hat{\lambda} = 0.5$ .  $\delta = 2$ .  $M \in \{1, 2, 4, 8\}$ .

case,  $\bar{T}^*$  monotonically increases with  $\hat{\lambda}$ , with  $\bar{T}^* \rightarrow 1 + \delta$  for  $\hat{\lambda} \rightarrow 0$  and  $\bar{T}^* \rightarrow \infty$  for  $\hat{\lambda} \rightarrow \hat{\lambda}_{max}$ . For  $M > 1$ , however,  $\bar{T}^* \rightarrow \infty$  for both  $\hat{\lambda} \rightarrow 0$  and  $\hat{\lambda} \rightarrow \hat{\lambda}_{max}$ . It has been shown in Section V-A that as the batch size  $M$  increases, both the maximum data throughput  $\hat{\lambda}_{max}$  and the unsaturated region of transmission probability  $S_{un}(q, \hat{\lambda})$  can be enlarged. For the minimum mean queueing delay of data packets  $\bar{T}^*$ , however, using a larger batch size  $M$  under a low aggregate input rate  $\hat{\lambda}$  may result in higher  $\bar{T}^*$  as the mean waiting time for forming a batch is significantly increased.

### C. Simulation Results

This section presents the simulation results to verify the preceding theoretical analysis. We consider an  $n$ -node connection-based slotted Aloha network where each node has Bernoulli packet arrivals with rate  $\lambda$ . The simulation setting is identical to the system model described in Section III. Each simulation is carried out for  $10^8$  time slots, and the initial number of data packets in each node's buffer is set to 20. In simulations, the steady-state probability of successful transmission of HOL requests  $p$  is obtained by calculating the ratio of the number of successful requests to the total number of transmitted requests; the data throughput  $\hat{\lambda}_{out}$  is obtained by calculating the ratio of the number of successful data packets to the total number of time slots  $10^8$ ; the mean queueing delay of data packets  $\bar{T}$  is obtained by calculating the ratio of the sum of queueing delay of all the arrival data packets to the total number of arrival data packets.

Section IV has demonstrated that the network operates at either the unsaturated steady-state point  $p_L$  or the saturated steady-state point  $p_A$ , where  $p_L$  is invariant with respect to the transmission probability  $q$  and determined by the batch size  $M$ , the aggregate input rate  $\hat{\lambda}$  and the overhead time for each successful transmission  $\delta$ , while  $p_A$  only depends on the transmission probability  $q$  and the number of nodes  $n$ . As Fig. 6a illustrates, with  $\hat{\lambda} = 0.5$ ,  $n = 100$  and  $\delta = 2$ , the unsaturated steady-state point  $p_L$  varies with the batch size  $M$  and is equal to 0.5384 for  $M = 4$  and 0.8443 for  $M = 8$ ; for

given  $M$ ,  $p_L$  does not change with the transmission probability  $q$ . The saturated steady-state point  $p_A$ , on the other hand, is closely dependent of  $q$  and independent of  $M$ .

Section V-A shows that the data throughput  $\hat{\lambda}_{out}$  is equal to the aggregate input rate  $\hat{\lambda}$  when the transmission probability  $q$  is selected from the unsaturated region  $S_{un}(q, \hat{\lambda})$ . With  $\hat{\lambda} \geq \hat{\lambda}_{max}$ ,  $S_{un}(q, \hat{\lambda}) = \emptyset$  and the network is always saturated for any  $q$ ; in this case, the maximum data throughput  $\hat{\lambda}_{max}$  can be achieved at the optimal transmission probability  $q^* = \frac{1}{n}$ . As shown in Fig. 6b, with the aggregate input rate  $\hat{\lambda} = 0.5$ , the maximum data throughput  $\hat{\lambda}_{max}$  is achieved at  $q^* = 0.01$  for  $M = 1$  and  $M = 2$ , i.e.,  $\hat{\lambda}_{max}^{M=1} = 0.21$  and  $\hat{\lambda}_{max}^{M=2} = 0.35$ ; for  $M = 4$  and  $M = 8$ , we have  $S_{un}^{M=4}(q, 0.5) = (0.0062, 0.0151)$  and  $S_{un}^{M=8}(q, 0.5) = (0.0017, 0.0307)$ , within which the network is unsaturated and the data throughput  $\hat{\lambda}_{out} = \hat{\lambda} = 0.5$ . Fig. 6b also corroborates that with a larger batch size  $M$ , the maximum data throughput  $\hat{\lambda}_{max}$  and the unsaturated region  $S_{un}(q, \hat{\lambda})$  are both enlarged.

The expression of the mean queueing delay of data packets  $\bar{T}$  with the transmission probability  $q \in S_{un}(q, \hat{\lambda})$  has been derived in Section V-B. As Fig. 7 illustrates, the minimum mean queueing delay  $\bar{T}^*$  is achieved at  $q_T^*$  given in (43). It can also be seen in Fig. 7a that for a smaller aggregate input rate, i.e.,  $\hat{\lambda} = 0.2$ , a larger batch size  $M$  leads to higher  $\bar{T}^*$  because data packets need to wait longer for forming a batch. On the other hand, with  $\hat{\lambda} = 0.5$ , we have  $\bar{T}_{M=4}^* > \bar{T}_{M=8}^*$ , as shown in Fig. 7b, indicating that the minimum mean queueing delay performance can be improved by appropriately increasing the batch size  $M$  under large aggregate input rates.

## VI. CONNECTION-FREE OR CONNECTION-BASED?

In contrast to connection-based random access, the basic concept of connection-free random access is that each node transmits its data packets without establishing connections. Without extra overhead time for connection establishment, connection-free random access has been widely argued to be more suitable for the emerging M2M

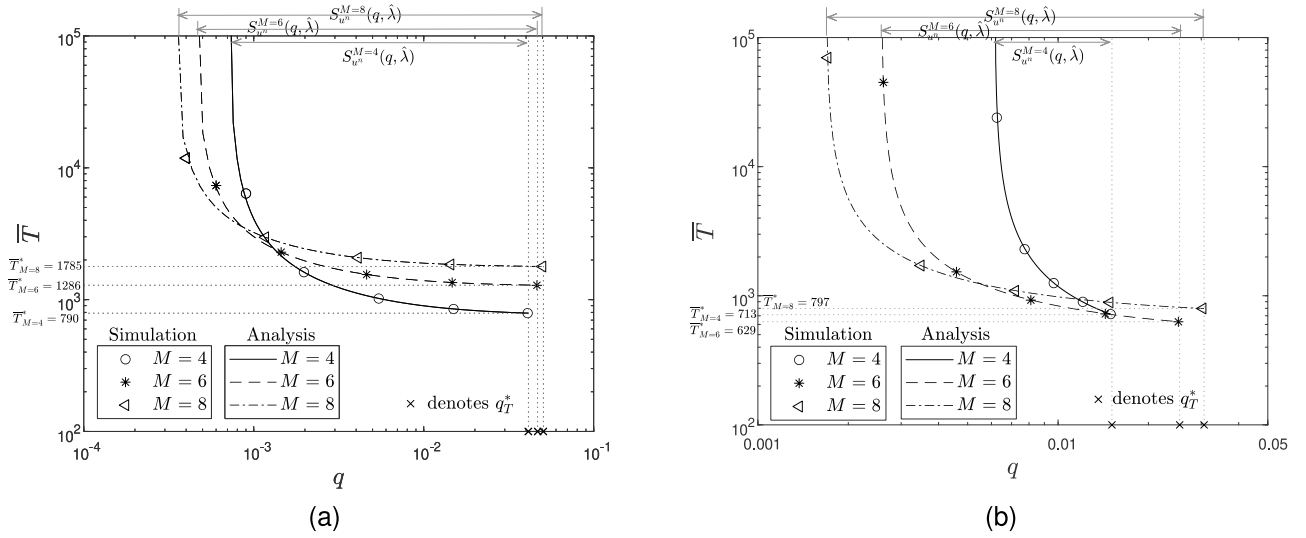


Fig. 7. Mean queueing delay  $\bar{T}$  versus transmission probability  $q$ .  $n = 100$ .  $\delta = 2$ .  $M \in \{4, 6, 8\}$ . (a)  $\hat{\lambda} = 0.2$ . (b)  $\hat{\lambda} = 0.5$ .

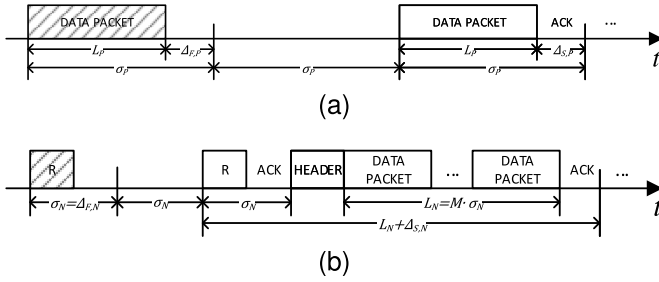


Fig. 8. Graphic illustration of (a) connection-free and (b) connection-based slotted Aloha.

communications [4], [5], [6], which is featured by massive MTDs with short packets and sporadic packet arrivals. However, each failed data packet transmission in connection-free random access usually wastes more time than each failed request transmission in connection-based random access. To see which one is more preferable for M2M communications, in this section, we conduct a comparative study between connection-free and connection-based slotted Aloha and apply the results to the 5G cellular networks.

For superscripts and subscripts throughout this section,  $P$  and  $N$  denote connection-free and connection-based, respectively.

#### A. Performance Comparison

Fig. 8 illustrates the connection-free and connection-based slotted Aloha. To be consistent with the notations in [20], we use  $\sigma$ ,  $L$ ,  $\Delta_S$  and  $\Delta_F$  to denote the time slot length, the time for each payload transmission, the overhead time for each successful or failed transmission, respectively, all in the unit of milliseconds (ms). As shown in Fig. 8a, with connection-free slotted Aloha, each successful or failed transmission takes one time slot, i.e.,  $L_P + \Delta_{S,P} = L_P + \Delta_{F,P} = \sigma_P$ . With connection-based slotted Aloha, as seen from Fig. 8b, each successful transmission includes  $M$  time slots for payload transmission and  $\delta$  time slots for signaling overhead, i.e.,

$L_N = M \cdot \sigma_N$  and  $\Delta_{S,N} = \delta \cdot \sigma_N$ ; each failed transmission takes one time slot, i.e.,  $\Delta_{F,N} = \sigma_N$ .

In the following subsections, we will compare the performance between connection-free and connection-based slotted Aloha in terms of maximum data throughput, unsaturated region of transmission probability, and minimum mean queueing delay of data packets.

1) *Maximum Data Throughput*: The maximum data throughput<sup>4</sup>  $\hat{\lambda}_{\max}^P$  of connection-free slotted Aloha is given by Eq. (10) in [20] as

$$\hat{\lambda}_{\max}^P = e^{-1} \cdot \frac{L_P}{L_P + \Delta_{S,P}}. \quad (44)$$

On the other hand, substituting  $\delta = \frac{\Delta_{S,N}}{\sigma_N}$ ,  $M = \frac{L_N}{\sigma_N}$  and  $\sigma_N = \Delta_{F,N}$  into (30), we can rewrite the maximum data throughput  $\hat{\lambda}_{\max}^N$  of connection-based slotted Aloha as

$$\hat{\lambda}_{\max}^N = \frac{L_N}{L_N + \Delta_{S,N} + (e-1) \cdot \Delta_{F,N}}. \quad (45)$$

Note that to achieve the maximum data throughput  $\hat{\lambda}_{\max}$ , the aggregate input rate  $\hat{\lambda}$  must be large enough to push all the queues to the saturated condition, and the transmission probability should be set to the optimal value  $q_{\lambda}^*$ .

By combining (44) and (45), we have  $\hat{\lambda}_{\max}^N > \hat{\lambda}_{\max}^P$  if and only if the payload lengths and the overhead parameters satisfy

$$\frac{L_N}{(e-1) \cdot \Delta_{F,N} + \Delta_{S,N}} > \frac{L_P}{(e-1) \cdot L_P + e \cdot \Delta_{S,P}}. \quad (46)$$

Note that when  $L_N = L_P = L$ , (46) is reduced to  $\frac{L}{\Delta_{S,N}} > \frac{1-e \cdot \frac{\Delta_{S,P}}{\Delta_{S,N}}}{e-1} + \frac{\Delta_{F,N}}{\Delta_{S,N}}$ , which is consistent with Eq. (21) in [20].

2) *Unsaturated Region of Transmission Probability  $q$* : According to [21], the unsaturated region of transmission probability  $q$  for connection-free slotted Aloha can be written as

$$S_{un}^P(q, \hat{\lambda}_P) = \left\{ q : -\frac{1}{n} \mathbb{W}_0(-\hat{\lambda}_P) < q < -\frac{1}{n} \mathbb{W}_{-1}(-\hat{\lambda}_P) \right\}. \quad (47)$$

<sup>4</sup>Note that the maximum data throughput was referred to as the maximum effective throughput in [20].

For connection-based slotted Aloha, the unsaturated region  $S_{u^n}^N(q, \hat{\lambda}_N)$  is given in (25).

To compare the unsaturated regions  $S_{u^n}^P(q, \hat{\lambda}_P)$  and  $S_{u^n}^N(q, \hat{\lambda}_N)$ , we need to specify the aggregate input rates  $\hat{\lambda}_P$  and  $\hat{\lambda}_N$ . Note that in the unsaturated condition, the data throughput is given by  $\hat{\lambda}_{out}^P = \hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}}$  and  $\hat{\lambda}_{out}^N = \hat{\lambda}_N$  for connection-free and connection-based slotted Aloha, respectively. For achieving equal data throughput  $\hat{\lambda}_{out}^P = \hat{\lambda}_{out}^N$ , we need  $\hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}} = \hat{\lambda}_N$ . It can be shown that for  $\hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}} = \hat{\lambda}_N < \hat{\lambda}_{max}^N$ ,  $S_{u^n}^P(q, \hat{\lambda}_P) \subset S_{u^n}^N(q, \hat{\lambda}_N)$  if (46) holds.<sup>5</sup> It indicates that if payload lengths and overhead parameters satisfy the condition given in (46), in addition to achieving a higher maximum data throughput in the saturated condition, connection-based slotted Aloha is also more robust for staying unsaturated thanks to a larger unsaturated region of transmission probability  $q$ .

3) *Minimum Mean Queueing Delay*: Based on the first and second moments of access delay derived in [21], the mean queueing delay of data packets of connection-free slotted Aloha with Bernoulli arrivals can be obtained as

$$\bar{T}_P = \frac{n - \hat{\lambda}_P}{nq \cdot \exp\{\mathbb{W}_0(-\hat{\lambda}_P)\} - \hat{\lambda}_P}, \quad (48)$$

for  $q \in S_{u^n}^P(q, \hat{\lambda}_P)$ . We can see from (48) that  $\bar{T}_P$  is a monotonic decreasing function of transmission probability  $q$ . The optimal transmission probability to minimize  $\bar{T}_P$  is then given by

$$q_T^{*,P} = -\frac{1}{n} \mathbb{W}_{-1}(-\hat{\lambda}_P) - \epsilon, \quad (49)$$

where  $\epsilon$  is a positive small value. For connection-based slotted Aloha, the mean queueing delay  $\bar{T}_N$  of data packets and the corresponding optimal transmission probability  $q_T^{*,N}$  are given in (42) and (43), respectively.

Note that the minimum mean queueing delay  $\bar{T}_P^*$  and  $\bar{T}_N^*$  are in the unit of time slots, but the time slot lengths  $\sigma_P$  and  $\sigma_N$ , in the unit of ms, are usually not equal. Let  $\tilde{T}^*$  denote the minimum mean queueing delay in the unit of ms. For connection-free and connection-based slotted Aloha,  $\tilde{T}_P^*$  and  $\tilde{T}_N^*$  are given by  $\bar{T}_P^* \cdot \sigma_P$  and  $\bar{T}_N^* \cdot \sigma_N$ , respectively, where  $\sigma_P = L_P + \Delta_{S,P}$  and  $\sigma_N = \Delta_{F,N}$ . Furthermore, for a fair comparison, assume that the aggregate input rates satisfy  $\hat{\lambda}_N = \hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}}$ , so that both have the same data throughput  $\hat{\lambda}_{out}^P = \hat{\lambda}_{out}^N$  in the unsaturated condition.

With light traffic, i.e.,  $\hat{\lambda}_P \rightarrow 0$ , we have  $\tilde{T}_P^* \rightarrow \sigma_P = L_P + \Delta_{S,P}$ , and  $\tilde{T}_N^* \rightarrow (1 + \delta) \cdot \sigma_N = \Delta_{F,N} + \Delta_{S,N}$  for  $M = 1$  or  $\tilde{T}_N^* \rightarrow \infty$  for  $M > 1$ , indicating that connection-free slotted Aloha may have lower minimum mean queueing delay if  $L_P + \Delta_{S,P} < \Delta_{F,N} + \Delta_{S,N}$ . As  $\hat{\lambda}_P$  increases, however,  $\tilde{T}_P^*$  increases faster as  $\tilde{T}_P^* = \infty$  for  $\hat{\lambda}_P \geq \hat{\lambda}_{max}^P$  but  $\tilde{T}_N^* < \infty$  for

<sup>5</sup>Specifically, by combining (47) and (25), we have  $S_{u^n}^P(q, \hat{\lambda}_P) \subset S_{u^n}^N(q, \hat{\lambda}_N)$  if and only if  $\frac{\hat{\lambda}_N}{M - \hat{\lambda}_N \cdot (M + \delta - 1)} < \hat{\lambda}_P$ . When (46) holds, for  $\hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}} = \hat{\lambda}_N < \hat{\lambda}_{max}^N$ , we have  $\frac{\hat{\lambda}_N}{\hat{\lambda}_P \cdot [M - \hat{\lambda}_N \cdot (M + \delta - 1)]} = \frac{L_P \cdot \Delta_{F,N}}{(L_P + \Delta_{S,P}) \cdot [L_N - \hat{\lambda}_N (L_N + \Delta_{S,N} - \Delta_{F,N})]} < \frac{L_P \cdot \Delta_{F,N}}{(L_P + \Delta_{S,P}) \cdot L_N} < 1$  by applying  $\delta = \frac{\Delta_{S,N}}{\Delta_{F,N}}$  and (45), leading to  $S_{u^n}^P(q, \hat{\lambda}_P) \subset S_{u^n}^N(q, \hat{\lambda}_N)$ .

$\hat{\lambda}_{max}^P \leq \hat{\lambda}_P < \hat{\lambda}_{max}^N \cdot \frac{L_P + \Delta_{S,P}}{L_P}$ . It can be seen that in addition to the overhead parameters and payload lengths, which one performs better in terms of the minimum mean queueing delay  $\tilde{T}^*$ , connection-free or connection-based, further depends on their data input rates.

### B. Case Study: Random Access in 5G Cellular Networks

To demonstrate how to apply the above analysis to practical networks, let us consider the 2-step RA-SDT and 4-step RA-SDT schemes specified in 3GPP Release 17 [22], as shown in Fig 9. Compared to the 4-step RA-SDT scheme, where each MTD has to first transmit Msg1 to obtain an uplink grant for Msg3 transmission, the 2-step RA-SDT scheme allows each MTD to transmit Msg1 and Msg3 in one message, MsgA, without an uplink grant. Thus, 2-step RA-SDT and 4-step RA-SDT are corresponding to grant-free random access and grant-based random access, respectively.

As Fig. 9a illustrates, with grant-free 2-step RA-SDT scheme, each node transmits its data payload along with the preamble in MsgA. If the Random Access Response (RAR) message and the Radio Resource Control (RRC) Release message are received within the MsgB Response Window, then the node knows that its MsgA transmission has been successful. With grant-based 4-step RA-SDT scheme, on the other hand, each node can transmit its data payload in Msg3 only after sending the preamble and receiving the RAR message within the Msg2 Response Window, as shown in Fig. 9b. The overhead parameters of 2-step and 4-step RA-SDT schemes are summarized in Table I.

Without loss of generality, we consider one single preamble. The time slot length is set to the interval between two consecutive Physical Random Access CHannel (PRACH) occasions available for preamble transmission of MTDs.<sup>6</sup> Table II lists the typical values of key system parameters [41], [42]. With the Transmission Time Interval (TTI) set to 0.5 ms, the overhead time for each successful and failed transmission in the grant-free 2-step RA-SDT and grant-based 4-step RA-SDT schemes can be obtained as

$$\Delta_{S,P} = \Delta_{F,P} = 6 \text{ ms} \quad \text{and} \quad \Delta_{S,N} = 8 \text{ ms}, \Delta_{F,N} = 2 \text{ ms}. \quad (50)$$

The payload transmission time  $L_P$  and  $L_N$  should be integer times of TTI. Given the system setting above, let us compare the performance of grant-free 2-step RA-SDT and grant-based 4-step RA-SDT in 5G cellular networks in terms of maximum data throughput, unsaturated region of transmission probability  $q$ ,<sup>7</sup> and minimum mean queueing delay in the following subsections.

<sup>6</sup>As [41] specifies, PRACH occasions for each MTD's preamble transmission are available periodically, that is, each MTD can start its MsgA or Msg1 transmission only at the beginning of each periodic PRACH occasion.

<sup>7</sup>Note that in 5G random access schemes, the transmission probability  $q$  of each request is determined by the Access Class Barring (ACB) factor  $q_{ACB}$  and the Uniform Backoff (UB) window size  $W_{UB}$  (in the unit of ms) [43]. Let  $\tilde{W}_{UB}$  denote the UB window size in the unit of time slots, which can be obtained from  $W_{UB} = \tilde{W}_{UB} \cdot \frac{W_{UB}}{\sigma}$ , with  $\sigma$  denoting the time slot length. Then the transmission probability  $q$  is given by  $\frac{2q_{ACB}}{\tilde{W}_{UB} + 1}$ .

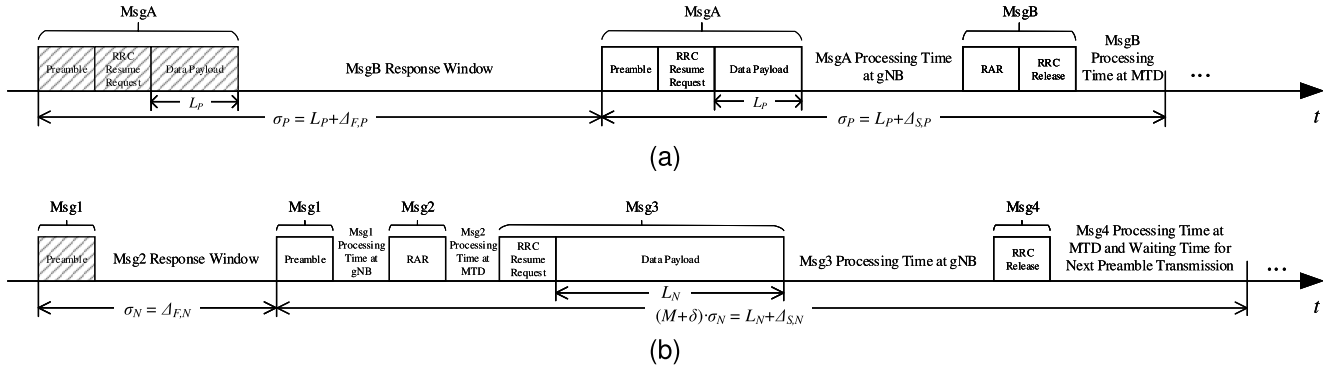


Fig. 9. Graphic illustration of (a) 2-step RA-SDT and (b) 4-step RA-SDT schemes in 5G cellular networks [22].

TABLE I  
OVERHEAD PARAMETERS OF 2-STEP RA-SDT AND 4-STEP RA-SDT SCHEMES

|                              |                                                                                                                                                                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grant-Free<br>2-Step RA-SDT  | $\Delta_{S,P}$ = Preamble Transmission Time + RRC Resume Request Transmission Time + MsgA Processing Time at gNB + RAR Transmission Time + RRC Release Transmission Time + MsgB Processing Time at MTD                                                                                                           |
|                              | $\Delta_{F,P}$ = Preamble Transmission Time + RRC Resume Request Transmission Time + MsgB Response Window Length                                                                                                                                                                                                 |
| Grant-Based<br>4-Step RA-SDT | $\Delta_{S,N}$ = Preamble Transmission Time + Msg1 Processing Time at gNB + RAR Transmission Time + Msg2 Processing Time at MTD + RRC Resume Request Transmission Time + Msg3 Processing Time at gNB + RRC Release Transmission Time + Msg4 Processing Time at MTD + Waiting Time for Next Preamble Transmission |
|                              | $\Delta_{F,N}$ = Preamble Transmission Time + Msg2 Response Window Length                                                                                                                                                                                                                                        |

TABLE II  
SYSTEM PARAMETER SETTING [41], [42]

|                              | Parameter                                                     | Value        |
|------------------------------|---------------------------------------------------------------|--------------|
| Grant-Free<br>2-Step RA-SDT  | MsgA Processing Time at gNB                                   | 1 TTI + 3 ms |
|                              | MsgB Processing Time at MTD                                   | 1 TTI        |
|                              | MsgB Response Window Length                                   | 5 ms         |
| Grant-Based<br>4-Step RA-SDT | Msg1 Processing Time at gNB                                   | 1 TTI        |
|                              | Msg2 Processing Time at MTD                                   | 1 TTI        |
|                              | Msg3 Processing Time at gNB                                   | 3 ms         |
|                              | Msg4 Processing Time at MTD                                   | 1 TTI        |
|                              | Waiting Time for Next Preamble Transmission                   | 1.5 ms       |
| Common Setting               | Msg2 Response Window Length                                   | 1.5 ms       |
|                              | Preamble/RRC Resume Request/RAR/RRC Release Transmission Time | 1 TTI        |

1) *Maximum Data Throughput*: With the overhead parameters given in (50), (46) can be simplified to

$$L_P < \frac{16.31}{11.44 - 1.72L_N} L_N, \quad (51)$$

with which the maximum data throughput with grant-based 4-step RA-SDT  $\hat{\lambda}_{\max}^N$  is larger than that with grant-free 2-step RA-SDT  $\hat{\lambda}_{\max}^P$ . When the payload length of grant-based 4-step RA-SDT  $L_N = 2$  ms, for instance, according to (51),  $\hat{\lambda}_{\max}^P < \hat{\lambda}_{\max}^N$  as long as  $L_P < 4.08$  ms. It implies that transmitting short data packets with grant-free 2-step RA-SDT would result in loss of maximum data throughput in the saturated condition.

2) *Unsaturated Region of Transmission Probability  $q$* : The unsaturated regions with grant-free 2-step RA-SDT  $S_{u^n}^P(q, \hat{\lambda}_P)$  and grant-based 4-step RA-SDT  $S_{u^n}^N(q, \hat{\lambda}_N)$  with  $\hat{\lambda}_N = \hat{\lambda}_P \cdot \frac{L_P}{L_P + \Delta_{S,P}} = \hat{\lambda}_P \cdot \frac{L_P}{L_P + 6}$  can be obtained by combining (47) and (25) with (50), respectively, which are illustrated in Fig. 10. Both shrink as the number of nodes  $n$  increases. With  $L_P < 4.08$  ms,  $S_{u^n}^P(q, \hat{\lambda}_P) \subset S_{u^n}^N(q, \hat{\lambda}_N)$ , indicating that in addition to a smaller maximum data throughput, the

system is also more prone to become saturated when grant-free 2-step RA-SDT is adopted.

3) *Minimum Mean Queueing Delay*: Given the overhead parameters in (50), we have  $\hat{T}_P^* = \bar{T}_P^* \cdot \sigma_P = \bar{T}_P^* \cdot (L_P + 6)$  ms and  $\hat{T}_N^* = \bar{T}_N^* \cdot \sigma_N = \bar{T}_N^* \cdot 2$  ms. The minimum mean queueing delay  $\bar{T}_P^*$  and  $\bar{T}_N^*$  are achieved at  $q_{T^*}^{*,P}$  given in (49) and  $q_{T^*}^{*,N}$  given in (43), respectively. Fig. 11 illustrates how the minimum mean queueing delay  $\hat{T}_P^*$  of grant-free 2-step RA-SDT and  $\hat{T}_N^*$  of grant-based 4-step RA-SDT vary with the aggregate data input rate  $\hat{\lambda}_N$  under different values of the number of nodes. Again  $\hat{\lambda}_N = \hat{\lambda}_P \cdot \frac{L_P}{L_P + 6}$  such that equal data throughput is achieved in both cases. It can be observed from Fig. 11 that even for very short data packets, e.g.,  $L_P = 0.5$  ms, grant-free 2-step RA-SDT may lead to longer minimum mean queueing delay. As Fig. 11a shows, grant-based 4-step RA-SDT outperforms grant-free 2-step RA-SDT with  $L_P = 0.5$  ms except for extremely small aggregate data input rate, i.e.,  $\hat{\lambda}_N < 3.8 \times 10^{-6}$ . The gain in the minimum mean queueing delay is even more significant as the number of MTDs increases from 20 to 100, as shown in Fig. 11b.

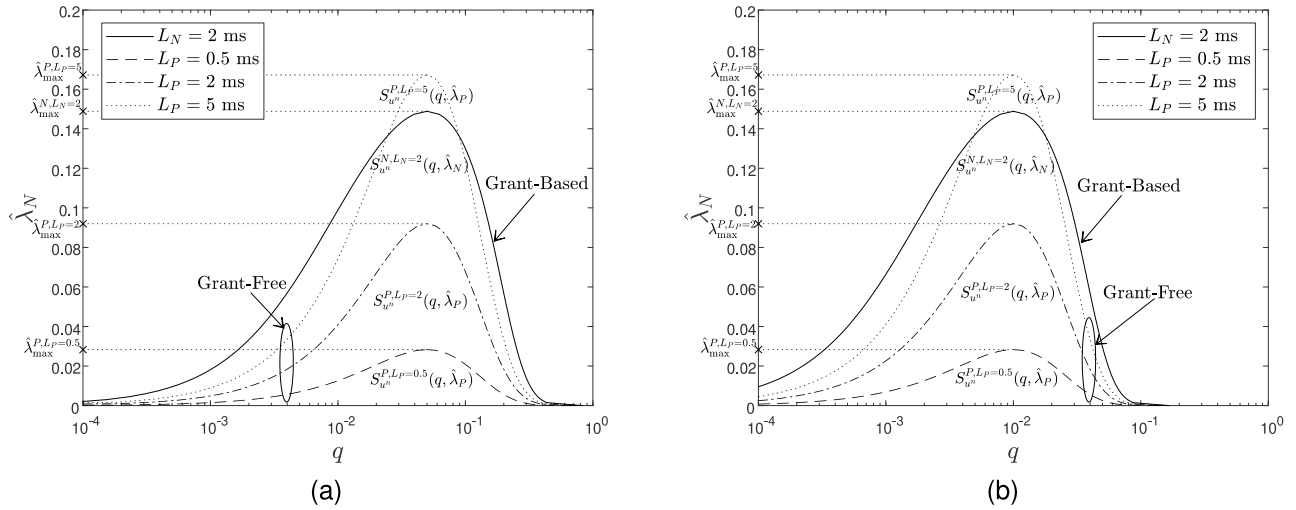


Fig. 10. Unsaturated regions of transmission probability  $S_u^P(q, \hat{\lambda}_P)$  and  $S_u^N(q, \hat{\lambda}_N)$  of grant-free 2-step RA-SDT and grant-based 4-step RA-SDT schemes in 5G cellular networks.  $\hat{\lambda}_N = \hat{\lambda}_P \cdot \frac{L_P}{L_P + 6}$ .  $L_N = 2$  ms.  $L_P \in \{0.5, 2, 5\}$  ms. (a)  $n = 20$ . (b)  $n = 100$ .

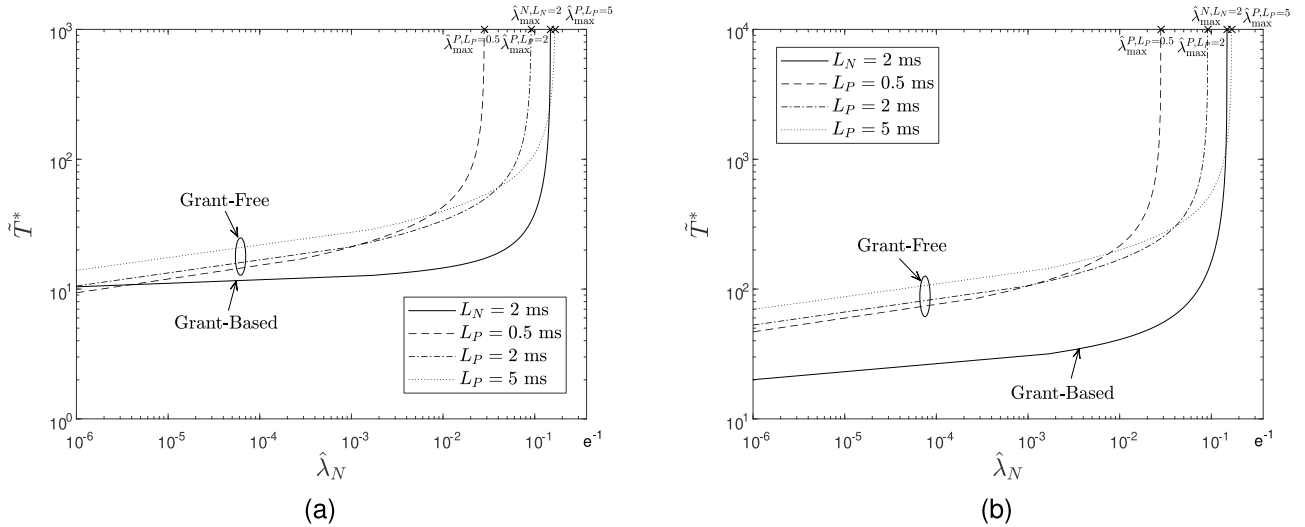


Fig. 11. Minimum mean queueing delay of data packets  $\tilde{T}_P^*$  and  $\tilde{T}_N^*$  in the unit of ms of grant-free 2-step RA-SDT and grant-based 4-step RA-SDT schemes in 5G cellular networks.  $\hat{\lambda}_N = \hat{\lambda}_P \cdot \frac{L_P}{L_P + 6}$ .  $L_N = 2$  ms.  $L_P \in \{0.5, 2, 5\}$  ms. (a)  $n = 20$ . (b)  $n = 100$ .

For M2M communications where a typical scenario is a large number of MTDs with short packets and low data input rate, we can conclude from the above results that in addition to higher maximum data throughput in the saturated condition and better robustness for staying unsaturated, grant-based 4-step RA-SDT may also achieve lower minimum mean queueing delay than grant-free 2-step RA-SDT.

## VII. DISCUSSIONS

Note that the above analysis is based on a few key assumptions. In this section, we will further discuss how to extend the analysis to incorporate: 1) asymmetric input rates and transmission probabilities of nodes, 2) finite buffer size, and 3) more advanced receiver design.

### A. Asymmetric Scenario

The preceding analysis has focused on the symmetric scenario, where all the nodes have the same data input rate

and transmission probability. With distinct applications and quality-of-service requirements of nodes, however, the input rate and transmission probability may vary from node to node. It is, therefore, of great practical importance to extend the analysis to the general asymmetric scenario. For connection-free Aloha, it has been shown in [44] that similar to the analysis for the symmetric case, the key to characterizing the service rates of nodes' queues still lies in: 1) establishing a Markov renewal process for each node's HOL packets, and 2) developing the fixed-point equations of steady-state probabilities of successful transmission of HOL packets. Different from the symmetric case, however, different nodes may have distinct steady-state probabilities of successful transmission of HOL packets, which need to be jointly solved.

For connection-based Aloha, the analytical framework proposed in this paper for the symmetric scenario can also be extended to the general asymmetric scenario. Specifically, for Node  $i \in \{1, \dots, n\}$ , a Markov renewal process can be

established for its HOL request, based on which the service rate of Node  $i$ 's queue can be derived as a function of  $p_i$ , the steady-state probability of successful transmission of Node  $i$ 's HOL requests conditioned on no data transmission. By establishing and solving the fixed-point equations of  $(p_1, \dots, p_n)$ , the throughput and queueing delay performance of each node can be analyzed by following a similar approach as presented in Section V. With nodes having distinct performance, however, fairness needs to be further taken into consideration when optimizing the system performance.

### B. Finite-Size Buffer

In the preceding analysis, each node is assumed to have an infinite-size data buffer. In practice, however, the buffer size is finite, with which arrival data packets would be dropped if there is no vacancy in the buffer. Intuitively, the optimal throughput performance is independent of the buffer size, as it is achieved only in the saturated condition with all nodes' queues always nonempty, and determined by the contention of nodes' HOL requests. The minimum mean queueing delay of data packets, on the other hand, could be improved by reducing the buffer size, because for the data packets that are not dropped, the waiting time in the buffer would be on average shorter with a smaller buffer size.

To extend the analysis to the finite-buffer-size scenario, the key still lies in the characterization of service processes of nodes' queues, which is determined by the aggregate activities of HOL requests. Specifically, we can adopt the methodology proposed in this paper to first model the behavior of HOL requests and then derive the steady-state probability of successful transmission of HOL requests. The throughput and delay performance of each data queue can be further characterized by following the standard finite-capacity queueing analysis.

### C. Advanced Receiver Design

Another key assumption in the preceding analysis is that the collision model is adopted at the receiver side. It is important to further incorporate more sophisticated receiver design into the analysis. To that end, the proposed Markov renewal process for HOL requests remains unchanged. Yet the fixed-point equation of the steady-state probability of successful transmission of HOL requests needs to be re-established, as it closely depends on the definition of successful transmission that varies under different assumptions of the receiver type and channel model. Specifically, (13) can be generalized to

$$p = \sum_{k=0}^{n-1} \binom{n-1}{k} (1-\omega)^{n-1-k} \omega^k \cdot \psi_k, \quad (52)$$

where  $\omega$  is the probability that a request queue is nonempty and its HOL request is transmitted conditioned on no data transmission, and  $\psi_k$  is the successful transmission probability of a HOL request for given  $k$  concurrent requests from other nodes. In this paper, with the collision receiver model over ideal channel conditions, we have  $\psi_0 = 1$  and  $\psi_k = 0$  for  $k > 0$ . For Rayleigh fading channels, i.e., received signal-to-noise ratios (SNRs) of nodes are exponentially distributed, the

conditional success probability  $\psi_k$  with the capture receiver<sup>8</sup> and the successive interference cancellation (SIC) receiver have been derived in [45] and [46], respectively. It can be expected that by adopting a more advanced receiver, the steady-state probability of successful transmission of HOL requests conditioned on no data transmission  $p$  can be improved, thus leading to gains in throughput and delay performance.

Note that with more sophisticated receivers, the transmission probability of each node can be jointly tuned with other system parameters when optimizing the network performance. For connection-free Aloha, it has been shown in [45] and [46] that with the capture receiver or SIC receiver, the SINR threshold is a key parameter that should be jointly tuned with the transmission probability of each node to maximize the sum data rate. For connection-based Aloha, the SINR threshold also determines a fundamental tradeoff between the transmission time of each request and the probability of successful transmission of requests: A larger SINR threshold leads to a higher encoding rate and thus shorter transmission time of a request, but lower probability of successful transmission of requests. As a result, by increasing the SINR threshold, the time slot length  $\sigma$ , which is determined by the transmission time of a request, can be reduced, yet it takes more time slots for a request to be successfully transmitted. To optimize the mean queueing delay of data packets, the SINR threshold and the transmission probability of each node should be jointly tuned.

## VIII. CONCLUSION

In this paper, the maximum data throughput and the minimum mean queueing delay of data packets for connection-based slotted Aloha are characterized and compared with those for connection-free slotted Aloha. Based on the proposed request-queue model, the service rate of each request queue is derived, from which the data throughput and the mean queueing delay of data packets are further obtained as functions of transmission probability of nodes  $q$ . As the mean queueing delay of data packets is finite only when the data queue is unsaturated, the region  $S_{un}(q, \hat{\lambda})$  of transmission probability  $q$  for each queue to stay unsaturated is also characterized. The analysis shows that to maximize the data throughput or to minimize the mean queueing delay, the optimal tuning of transmission probability  $q$  only requires the long-term statistical information such as the aggregate data input rate  $\hat{\lambda}$ . Moreover, with a larger batch size (or data payload transmission time)  $M$ , the maximum data throughput and unsaturated region of transmission probability can always be improved. Yet the minimum mean queueing delay could be increased when the data input rate is low.

The optimal performance with connection-based slotted Aloha is further compared to that with connection-free slotted Aloha from the aspects of maximum data throughput, unsaturated region of transmission probability, and minimum

<sup>8</sup>Specifically, with the capture receiver, a request can be successfully decoded as long as its received signal-to-interference-plus-noise ratio (SINR) is above a certain threshold.

mean queueing delay. For the first two, the condition that the payload lengths and overhead parameters should satisfy for beneficial connection establishment is obtained, which includes the payload length threshold derived in [20] as a special case. For the optimal queueing delay performance, on the other hand, whether gains can be brought by the connection establishment further depends on the data input rate of nodes. To demonstrate the practical insights of the analysis, a case study on grant-based 4-step RA-SDT and grant-free 2-step RA-SDT of 5G networks is also conducted. The comparison shows that with the current system setting, in addition to higher maximum data throughput and better robustness for staying unsaturated, the grant-based 4-step RA-SDT also leads to lower minimum mean queueing delay even when the aggregate data input rate is small, and the gain becomes more significant as the number of nodes increases.

#### APPENDIX A DERIVATION OF (11)-(12)

To derive the steady-state probability  $\tilde{\alpha}$  of no data transmission for a given HOL request, let us first denote  $E_1$  as the event of no data transmission, and  $E_2$  as the event of the HOL request in State 0 or the first time slot of State T. Define  $\alpha$  as the unconditional steady-state probability of no data transmission, which can be expressed as  $\alpha = \Pr\{E_1\}$ . We have

$$\tilde{\alpha} = \Pr\{E_1|E_2\} = \frac{\Pr\{E_2|E_1\}}{\Pr\{E_2\}} \cdot \alpha, \quad (53)$$

where

$$\Pr\{E_2\} = \rho(\tilde{\pi}_0 + \frac{\tilde{\pi}_T}{\tau_T}), \quad (54)$$

with  $\rho$ ,  $\tau_T$ ,  $\tilde{\pi}_0$  and  $\tilde{\pi}_T$  given in (14), (7) and (9), respectively. Given that there is no data transmission, each request queue is either empty or nonempty with a HOL request either in State 0 or the first time slot of State T. Consequently,  $\Pr\{E_2|E_1\}$  can be written as

$$\Pr\{E_2|E_1\} = \frac{\rho(\tilde{\pi}_0 + \frac{\tilde{\pi}_T}{\tau_T})}{1 - \rho + \rho(\tilde{\pi}_0 + \frac{\tilde{\pi}_T}{\tau_T})}. \quad (55)$$

By combining (7), (9)-(10), (14), and (53)-(55),  $\tilde{\alpha}$  can be obtained as (11).

To derive the unconditional steady-state probability of no data transmission  $\alpha$ , first note that  $M + \delta - 1$  time slots are reserved for data transmission after each successful request transmission. The probability that there is a successful request transmission at any time slot is equal to  $n\lambda_r$  and  $n\mu_r$  in the unsaturated and saturated conditions, respectively. We then have

$$\alpha = \begin{cases} 1 - n\lambda_r(M + \delta - 1) & \lambda_r < \mu_r \\ 1 - n\mu_r(M + \delta - 1) & \lambda_r \geq \mu_r. \end{cases} \quad (56)$$

Finally, the unconditional steady-state probability of no data transmission  $\alpha$  can be derived as (12) by combining (10)-(11) and (56).

#### APPENDIX B DERIVATION OF (37)-(38) AND (40)-(41)

1) *Derivation of (37)-(38)*: To derive the mean waiting time of each request  $\bar{W}_r$  in the unsaturated condition, we first introduce the number of arrival data packets during the waiting time and the service time of a request,  $N^+$ , and its mean,  $\bar{N}^+$ .  $\bar{W}_r$  can then be written as

$$\bar{W}_r = \frac{\bar{N}^+}{\lambda} - \bar{D}_r = \frac{\bar{N}^+}{\lambda} - \frac{1}{p_L \tilde{\alpha}_u q} - \tau_T + 1, \quad (57)$$

where  $\bar{D}_r$  is the mean service time of requests given in (36),  $p_L$  is the unsaturated steady-state point given in (18),  $\tilde{\alpha}_u = \frac{1 - n\lambda_r(M + \delta - 1)}{1 - \lambda_r(M + \delta - 1)}$  is the steady-state probability of no data transmission for a given HOL request in the unsaturated condition, and  $\tau_T = M + \delta$  is the time for each successful transmission.

[47] provided the analytical results for the  $E_k/G/1$  queue (i.e., the continuous-time counterpart of the  $NB_k/G/1$  queue), from which the probability generating function of  $N^+$  can be obtained as

$$G_{N^+}(z) = \frac{M \left(1 - \frac{\lambda_r}{\mu_r}\right) (z-1)}{\frac{z^M}{G_K(z)} - 1} \cdot \prod_{i=1}^{M-1} \frac{z - z_i}{1 - z_i}, \quad (58)$$

where  $z_1, \dots, z_{M-1}$  are the  $M - 1$  roots of  $z^M - G_K(z) = 0$  inside the unit circle  $|z| = 1$ , and  $G_K(z) = \sum_{j=0}^{\infty} \Pr\{K = j\} \cdot z^j$  is the probability generating function of  $K$ , the number of arrival data packets during the service time of a request. With Bernoulli arrivals of data packets,  $G_K(z)$  can be obtained from (35) as

$$G_K(z) = \sum_{j=0}^{\infty} \sum_{i=j}^{\infty} \binom{i}{j} \lambda^j (1 - \lambda)^{i-j} \Pr\{D_r = i\} z^j \\ = \frac{p_L \tilde{\alpha}_u q (\lambda z + 1 - \lambda)^{\tau_T}}{1 - (1 - p_L \tilde{\alpha}_u q) (\lambda z + 1 - \lambda)}. \quad (59)$$

By combining (58) and (59), and applying  $\lambda_r = \frac{\lambda}{M}$  and  $\mu_r = \frac{p_L \tilde{\alpha}_u q}{p_L \tilde{\alpha}_u q (\tau_T - 1) + 1}$  in the unsaturated condition,  $G_{N^+}(z)$  can be further derived as

$$G_{N^+}(z) = \frac{[M - \lambda \left(\frac{1}{p_L \tilde{\alpha}_u q} + \tau_T - 1\right)] (z-1) (\lambda z + 1 - \lambda)^{\tau_T}}{z^M \left[1 + \lambda \left(\frac{1}{p_L \tilde{\alpha}_u q} - 1\right) (1-z)\right] - (\lambda z + 1 - \lambda)^{\tau_T}} \cdot \prod_{i=1}^{M-1} \frac{z - z_i}{1 - z_i} \\ = (\lambda z + 1 - \lambda)^{\tau_T} \prod_{i=1}^{\delta} \frac{1 - r_i}{1 - r_i z}, \quad (60)$$

where  $z_1, \dots, z_{M-1}$  are the  $M - 1$  roots inside the unit circle  $|z| = 1$  of the equation  $z^M - G_K(z) = 0$ , which can be rewritten from (59) as

$$z^M - \frac{p_L \tilde{\alpha}_u q (\lambda z + 1 - \lambda)^{\tau_T}}{1 - (1 - p_L \tilde{\alpha}_u q) (\lambda z + 1 - \lambda)} = 0. \quad (61)$$

$r_1, \dots, r_{\delta}$  are the roots inside the unit circle  $|r| = 1$  of Eq. (38) of  $r$ , which is obtained by replacing  $z$  in (61) with  $\frac{1}{r}$ . The average number of arrival data packets during the waiting time and the service time of each request,  $\bar{N}^+$ , can then be derived from (60) as

$$\bar{N}^+ = G_{N^+}^{(1)}(1) = \lambda \tau_T + \sum_{i=1}^{\delta} \frac{r_i}{1 - r_i}. \quad (62)$$

By substituting (62) into (57) and applying  $\tau_T = M + \delta$ , the mean waiting time of each request  $\bar{W}_r$  can be finally obtained as (37).

2) *Derivation of (40)-(41)*: By ignoring  $\tau_T - 1$  time slots in each service time, each request queue can be approximated as an  $NB_k/Geo/1$  queue where the service time follows the Geometric distribution with parameter  $p_L \tilde{\alpha}_u q$ . By replacing  $\tau_T$  in (57) with 1, the mean waiting time of each request can be approximated as

$$\bar{W}_r \approx \frac{\tilde{N}^+}{\lambda} - \frac{1}{p_L \tilde{\alpha}_u q}, \quad (63)$$

where  $\tilde{N}^+$  is the average number of arrival data packets during the waiting time and the service time of each request in the  $NB_k/Geo/1$  queue. By replacing  $\tau_T$  in (60) with 1, the probability generating function of  $N^+$  of the  $NB_k/Geo/1$  queue can be obtained as

$$\tilde{G}_{N^+}(z) = (\lambda z + 1 - \lambda) \cdot \frac{1-r_0}{1-r_0 z}, \quad (64)$$

where  $r_0$  is the single real root in  $(0, 1)$  of Eq. (41) of  $r$ , which is obtained by replacing  $\tau_T$  with 1 and  $z$  with  $\frac{1}{r}$  in (61). The average number of arrival data packets during the waiting time and the service time of each request in the  $NB_k/Geo/1$  queue,  $\tilde{N}^+$ , can then be derived from (64) as

$$\tilde{N}^+ = \tilde{G}_{N^+}^{(1)}(1) = \lambda + \frac{r_0}{1-r_0}. \quad (65)$$

By combining (63) and (65), the mean waiting time of each request  $\bar{W}_r$  can be approximated as (40).

#### APPENDIX C MONOTONIC PROPERTY OF $\bar{T}$ GIVEN IN (42) WITH RESPECT TO $q$

We can see from (42) that the mean queueing delay  $\bar{T}$  monotonically increases with  $r_0$ , the root in  $(0, 1)$  of Eq. (41).  $r_0$  can be expressed as an implicit function of  $q$ , denoted as  $g(q)$ . We then rewrite (41) as

$$\sum_{i=1}^M g^i(q) - \lambda \cdot g^M(q) - \lambda \cdot \left( \frac{1}{p_L \tilde{\alpha}_u q} - 1 \right) = 0, \quad (66)$$

where  $p_L$  and  $\tilde{\alpha}_u$  are both independent of  $q$  in the unsaturated condition. For (66), taking the first derivative with respect to  $q$ , we further have

$$g^{(1)}(q) = - \frac{\lambda}{p_L \tilde{\alpha}_u q^2 [\sum_{i=1}^{M-1} i \cdot g^{i-1}(q) + M(1-\lambda) \cdot g^{M-1}(q)]}. \quad (67)$$

It is clear from (67) that  $g^{(1)}(q) < 0$  for  $g(q) \in (0, 1)$ , indicating that  $r_0 = g(q)$  monotonically decreases with  $q$ . As a result, we can conclude that the mean queueing delay  $\bar{T}$  given in (42) is a monotonic decreasing function of the transmission probability  $q$  in the unsaturated condition.

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